

Project Documentation: Data Cleaning & Preparation

Automated pipeline for processing transactional e-commerce data

Project Name:	Data Analytics Project 1
Environment:	Jupyter Notebook / Google Colab (Python 3)
Libraries Used:	pandasnumpy
Output Artifact:	Cleaned_Dataset.xlsx

1. Executive Summary & Goal

The objective of this project is to build a robust data cleaning and preparation workflow using Python to convert a raw, uncleaned retail dataset into a reliable, analytical-ready asset. Raw transactional datasets frequently suffer from operational anomalies such as missing data, formatting discrepancies, un-stripped strings, and floating-point variances. This documentation systematically covers every pipeline step taken within the codebase to identify, clean, and validate the dataset.

2. Dataset Overview & Dimensions

Initial exploratory data analysis (EDA) was performed to understand the structure, scale, and specific inconsistencies within the data.

1,200	14	309
TOTAL ROWS	TOTAL COLUMNS	NULL VALUES FIXED

Dataset Schema & Columns Identified:

- OrderID:** Unique transactional identifier
- Date:** Purchase date and time stamp
- CustomerID:** Unique client identifier
- Product:** Item name (e.g., Monitor, Phone, Desk)
- Quantity:** Number of items purchased
- UnitPrice:** Price per item unit
- ShippingAddress:** Delivery destination

8. **PaymentMethod:** Method used for payment
9. **OrderStatus:** Current fulfillment status (e.g., Shipped, Cancelled)
10. **TrackingNumber:** Courier tracking alphanumeric code
11. **ItemsInCart:** Total count of items in user session
12. **CouponCode:** Marketing promotional code applied
13. **ReferralSource:** Origin of user visit (e.g., Instagram, Email)
14. **TotalPrice:** Computed final transaction value

3. Data Cleaning Methodology & Pipeline

The following sequential operations were implemented programmatically to clean the source dataset:

Step 3.1: Missing / Null Value Resolution

An evaluation using `df.isna().sum()` showed that the `CouponCode` column had exactly **309 missing values**. Since a missing value in this context implies that no discount coupon was applied during checkout, the null values were filled with a descriptive string:

```
df['CouponCode'] = df['CouponCode'].fillna('Not Available(NA)')
```

Step 3.2: Formatting & Type Conversion (Temporal Data)

The `Date` field contained mixed/string characters. To ensure chronological sorting and time-series reliability, it was parsed into uniform `datetime64[ns]` objects:

```
df["Date"] = pd.to_datetime(df["Date"])
```

Step 3.3: Text Normalization & Whitespace Trimming

String values often contain accidental leading or trailing white spaces that can break categorical grouping and filters. All text-based columns were explicitly cast and normalized:

```
text_columns = ["Product", "ShippingAddress", "PaymentMethod", "OrderStatus",  
                "TrackingNumber", "CouponCode", "ReferralSource"]  
for col in text_columns:  
    df[col] = df[col].astype(str).str.strip()
```

Step 3.4: Numerical Precision Adjustment

Floating-point numbers in pricing metrics possessed inconsistent fractional digits. To improve clarity and standardize currency representations, all `float64` columns were systematically rounded to a single decimal place:

```
float_cols = df.select_dtypes(include = "float64").columns
df[float_cols] = df[float_cols].round(1)
```

4. Integrity Validation & Audit

Audit Metric	Raw State	Cleaned State	Status
Missing Values (CouponCode)	309 Nulls	0 Nulls (Filled with 'Not Available(NA)')	Resolved
String Padding/Spaces	Present in text values	Fully stripped (.str.strip())	Resolved
Date Datatype	Object / Text	Datetime64[ns]	Resolved
Float Precision	Unbounded decimals	Rounded uniformly to 1 decimal place	Resolved
Row Integrity	1,200 rows	1,200 rows (No valid rows dropped)	Validated

5. Conclusion

The cleaning script successfully standardized the data without loss of structural rows. The resulting artifact, `Cleaned_Dataset.xlsx`, is now structurally sound, normalized, and optimally prepared for downstream reporting, business intelligence dashboards, or machine learning applications.